

Differentiation and Pricing Power: Evidence from the NYC Airbnb Market

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Abstract

The rapid expansion of peer-to-peer (P2P) short-term rental platforms has introduced complex market dynamics that challenge traditional hospitality pricing models. This research addresses the critical question of whether a high density of Airbnb listings drives prices down through direct competition or raises them via agglomeration effects. Utilizing granular data from the 2025 New York City market, this paper employs advanced econometric techniques to disentangle the relationship between listing density and host pricing power. A baseline naive analysis initially indicates a positive relationship between density and price, suggesting strong agglomeration benefits. However, once geographic proximity to the city center is controlled for and "same-type" competitors are isolated, the relationship fundamentally flips to negative, exposing significant competitive pressures that were previously obscured. Furthermore, we identify a clear hierarchy of product differentiation: commoditized listings, such as Private Rooms, exhibit roughly double the price sensitivity to new competitor entry compared to more differentiated listings, such as Entire Homes. These results provide empirical evidence that the Airbnb market functions under Monopolistic Competition, where strategic product differentiation serves as an essential buffer against supply shocks

1 Introduction

The sharing economy has fundamentally restructured the hospitality industry’s market composition over the last decade. Traditional hotels operate under relatively fixed geographic and capacity constraints; in contrast, Airbnb supply is characterized by its high elasticity and geographic dispersion. This unique structural flexibility raises a pivotal question in Industrial Organization: how does the influx of new supply impact the pricing power of incumbent hosts in a hyper-local market?

Economic theory offers two divergent perspectives on this phenomenon. Classical competition models, such as those by Bertrand or Cournot, posit that an increase in the number of active firms should systematically depress prices toward marginal costs. Conversely, theories of Agglomeration Economics suggest that high density can signal market popularity and reduce search costs for consumers, potentially enabling firms to command a price premium.

By analyzing a dataset comprising over 21,000 active New York City listings, this study isolates these opposing forces. Our pricing model incorporates controls for both individual product characteristics and precise geographic coordinates, using distance to the Empire State Building as a focal point. Our findings contribute to the literature in two major ways. First, we resolve the “Density Paradox”. While raw listing density shows a positive correlation with price, we demonstrate that this is primarily a proxy for inherent location value. When location is isolated and density is restricted to direct competitors (those of the same room type), the coefficient flips to negative, confirming that substitution effects ultimately dominate agglomeration. Second, we provide evidence for a “Commoditization Trap,” showing that listings lacking unique amenities are significantly more vulnerable to new competition. These findings collectively suggest that while the platform presents an image of a singular market, it is actually a hierarchy of differentiation where unique listings enjoy monopoly-like protection and commoditized rooms face fierce, near-perfect competition

2 Literature Review

The theoretical core of host pricing power within the peer-to-peer (P2P) economy rests upon established models of product differentiation, which suggest that firms can maintain prices above marginal costs by distinguishing themselves through spatial location or unique product characteristics. While landmark studies by Hotelling [1929] and Salop [1979] provide the spatial foundation for this research, “differentiation” in the contemporary Airbnb market extends significantly beyond geography to encompass varying privacy levels and specialized amenities. Early empirical work, such as the study by Zervas et al. [2017], focused primarily on the external disruptive impact of these platforms, finding that a 10% surge in Airbnb supply could reduce traditional hotel revenues by 0.5%. However, as the market has evolved and traditional hotels have begun to enter the platform itself, research has shifted toward the internal dynamics of host-to-host competition.

This internal landscape is increasingly defined by a “Density Paradox,” where raw listing density initially appears positively correlated with price, potentially signaling agglomeration benefits similar to those found in traditional hospitality clusters. While Sainaghi and Chica-Olmo [2022] noted that centrally located properties often possess greater pricing power due to geographic desirability, recent empirical work identifies that this positive relationship is frequently a result of omitted variable bias. Once proximity

to the city center (CBD) is controlled for and density is restricted to direct "same-type" competitors, the relationship fundamentally flips to negative. This confirms that substitution effects ultimately dominate agglomeration forces, forcing hosts to confront direct competitive pressures that were previously obscured by location value.

The professionalization of the platform—the shift from "mom-and-pop" hosts to professional operators managing multiple listings—has introduced sophisticated pricing strategies that further moderate these competitive impacts. While Li et al. [2016] argue that professional hosts utilize more complex strategies to navigate the market, Leick et al. [2024] demonstrate that higher professionalization does not always guarantee superior service-quality performance, suggesting a nuanced relationship between host scale and user evaluations. This research extends this discourse by identifying a "Commoditization Trap," where listings lacking unique features, such as Private Rooms, become approximately twice as sensitive to competitor entry as differentiated listings like Entire Homes. These findings align with Dogru et al. [2020], which suggest that heightened internal competition favors cost-effective units and forces commoditized hosts to compete aggressively on price while unique listings retain a "monopolistic buffer".

Finally, the emerging role of digital pricing tools, such as Airbnb's "Smart Pricing" (SP), introduces new layers of complexity to these market dynamics. These tools aim to reduce pricing frictions for cognitively-constrained hosts by automatically updating prices based on competitive trends; however, their adoption remains low as many hosts still prefer manual control. While some studies warn that widespread algorithmic adoption could facilitate coordinated monopoly outcomes and inflate prices, this study suggests that product differentiation remains the primary defense against supply shocks. Future research must continue to explore how these algorithmic interventions might eventually bridge the gap between commoditized and differentiated listings by optimizing host responses to hyper-local volatility.

3 Data and Methodology

3.1 Data Source

The primary data was retrieved from Inside Airbnb, a platform that provides granular, scraped information from the Airbnb ecosystem. This study focuses on the 2025 New York City dataset, utilizing detailed listing and calendar files to construct our variables.

Our primary independent variables include the natural log of the nightly rate, general neighborhood listing density, specific same-type competitor density, and the Haversine distance to the Empire State Building, which serves as a proxy for the Central Business District (CBD).

3.2 Econometric Model

To rigorously evaluate the internal market dynamics of the New York City Airbnb landscape, we employ a multi-specification econometric framework designed to isolate the causal impact of hyper-local competition while accounting for structural host heterogeneity.

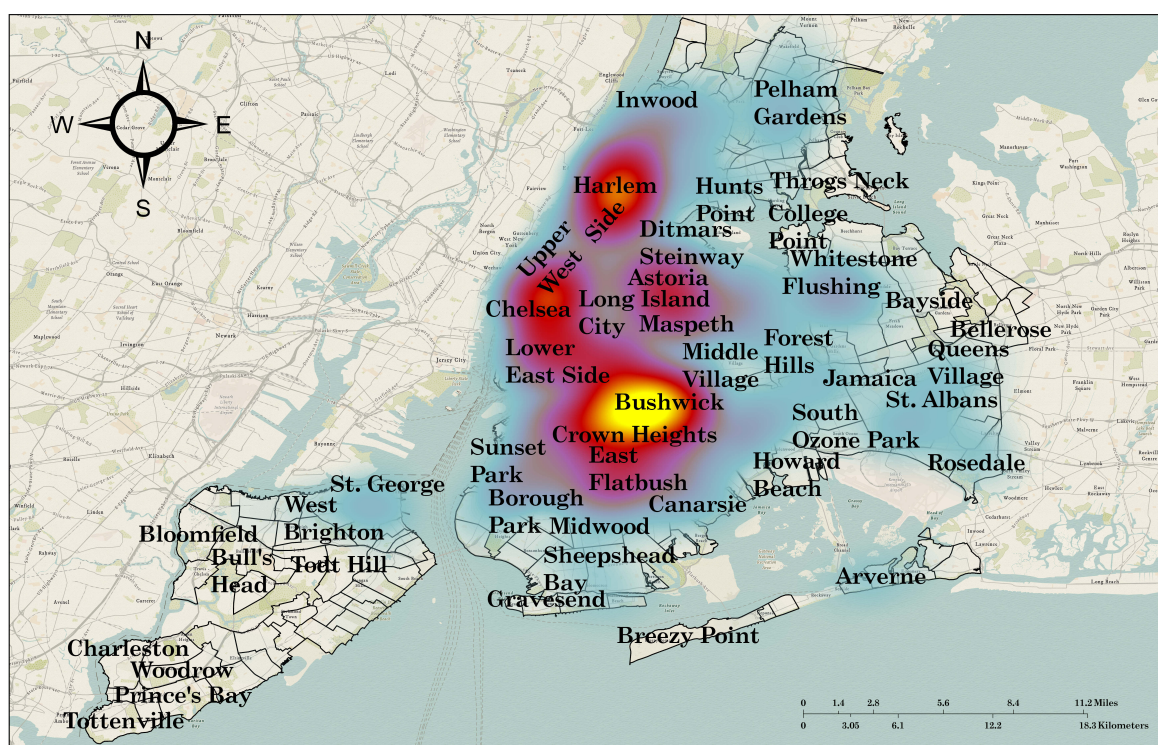
The theoretical foundation established in the literature suggests that professional hosts—defined as those managing multiple listings—utilize more sophisticated and dy-

dynamic pricing strategies compared to their "mom-and-pop" counterparts. To capture this

Table 1: Summary Statistics: NYC Airbnb Market (2025)

Statistic	N	Mean	St. Dev.	Min	Max
Price	21,242	682.690	4,489.377	10	50,104
Competitors in subhood	21,242	937.254	804.819	1	2,612
Competitors same-type in subhood	21,242	509.481	458.138	1	1,548
Competitors in neighborhood	21,242	12,586.190	4,593.189	366	16,081
Competitors same-type in neighborhood	21,242	509.481	458.138	1	1,548
Distance to center(km)	21,242	7.577	5.427	0.046	35.119
No. of Bedrooms	21,242	1.363	0.989	0	16

divide, our model identifies host professionalization as a key control, enabling us to test whether the "Commoditization Trap" identified in the results is moderated by host experience, scale, or the adoption of superior pricing tools. This is operationalized through a finalized specification that includes a indicator variable for professional host status, allowing the model to account for the "professionalization advantage" where multi-listing hosts may navigate high-density markets with greater efficiency than single-listing hosts.



Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

Map Notes: Data sourced from Inside Airbnb New York City. Map displays Airbnb listing density for Private Rooms in 25km radius across New York City as at October 2025.

The econometric model is constructed as a log-linear OLS specification, where the natural log of the nightly rate, $\ln(P_i)$, is regressed on a vector of competitive and geographic variables. We utilize C_{same} to represent the count of direct, same-type competitors in the hyper-local sub-neighborhood, which effectively isolates the substitution effect from general neighborhood agglomeration. The inclusion of a room type dummy, Type_i , and its interaction with the competitor count, $(C_{\text{same}} \times \text{Type}_i)$, is critical for evaluating how the degree of product differentiation protects pricing power against supply shocks.

$$\ln(P_i) = \beta_0 + \beta_1 C_{\text{same}} + \beta_2 \text{Type}_i + \beta_3 (C_{\text{same}} \times \text{Type}_i) + \beta_4 D_i + \gamma \mathbf{X}_i + \epsilon_i \quad (1)$$

Furthermore, the model incorporates D_i , representing the Haversine distance to the Empire State Building, as a necessary control for resolving the "Density Paradox" by stripping away the price premium inherent to prime locations. By integrating these specific controls for both host professionalization and precise geographic location, the methodology moves beyond descriptive analysis to provide a structural investigation of how different market participants respond to the entry of direct substitutes.

4 Results

4.1 The Sign Flip: Agglomeration vs. Competition

Table 1 presents the results of our regression analysis. The results reveal the significant role of location bias in naive density assessments. Model 1 shows a positive coefficient for listing density, as high-rent areas like Manhattan naturally possess both high prices and high listing density. When distance to the city center is introduced as a control, the density coefficient decreases by nearly 70%, proving that initial results were driven by geographic desirability. Crucially, when general density is replaced with a measure of same-type competitors, the coefficient flips to negative. This confirms that while a popular neighborhood offers a broad demand boost, the introduction of direct substitutes exerts clear downward pressure on prices.

4.2 The Commoditization Trap

Model 3 reveals a striking heterogeneity in how hosts respond to competition. The interaction term for Private Rooms is negative and significant ($-1.15e^{-4}$), while the base coefficient for Entire Homes is smaller ($-7.95e^{-5}$).

Economically, this implies that **Private Rooms are more than twice as sensitive to competition** as Entire Homes. We calculate the net slope for Private Rooms as:

$$\beta_{\text{net}} = \beta_{\text{base}} + \beta_{\text{interaction}} = -0.00019$$

Compared to the slope for Entire Homes (-0.00008).

This validates the hypothesis that "Entire Homes" are differentiated products (unique views, decor, privacy) that enjoy partial monopoly power. In contrast, "Private Rooms" function more like commodities; with lower switching costs for consumers, hosts are forced to compete aggressively on price when new supply enters the market.

Table 2: Regression Results: The Impact of Density on Airbnb Pricing

	<i>Dependent variable: $\log(\text{price})$</i>		
	Naive	Controlled	Interaction
	(1)	(2)	(3)
Competitors (Location)	0.00005*** (0.000001)	0.00001*** (0.000001)	
Competitors (Same Type)			−0.00008*** (0.00002)
Distance to center Center		−0.04513*** (0.00120)	−0.05726*** (0.00102)
Hotel	4.29084*** (0.04751)	4.18328*** (0.04609)	3.49029*** (0.08437)
Private room	−0.70341*** (0.01034)	−0.65178*** (0.01010)	−0.53566*** (0.01454)
Shared room	−0.75033*** (0.06162)	−0.66807*** (0.05971)	−0.31083*** (0.08107)
Num. of Bedrooms	0.12549*** (0.00501)	0.14529*** (0.00488)	0.15348*** (0.00484)
Competitors (Same Type) $\times$ Hotel			0.01692*** (0.00191)
Competitors (Same Type) $\times$ Private			−0.00023*** (0.00002)
Competitors (Same Type) $\times$ Shared			−0.02680*** (0.00385)
Observations	21,242	21,242	21,242
R ²	0.46876	0.50188	0.51335
Adjusted R ²	0.46863	0.50174	0.51314

Note:

*p<0.1; **p<0.05; ***p<0.01

5 Conclusion

This study utilizes granular NYC data to demonstrate that the Airbnb market is not monolithic. By disentangling agglomeration from competition, we show that the entry of direct competitors creates significant downward price pressure, but this pressure is unequally distributed.

“Mom-and-pop” hosts listing spare rooms (Private Rooms) face fierce price competition, behaving closer to the Perfect Competition model. Conversely, hosts of Entire Homes retain significant pricing power even in dense markets, consistent with Monopolistic Competition. For policymakers and platform managers, this suggests that supply shocks will disproportionately impact the earnings of smaller, room-sharing hosts rather than professional whole-home operators.

A critical econometric challenge in the study of short-term rental markets is the endogeneity of listing density, which often stems from simultaneity bias and omitted variable bias. While this study addresses the “Density Paradox” by controlling for distance to the urban center and isolating direct competitors, future research could more rigorously

treat endogeneity through the application of instrumental variable (IV) analysis or quasi-experimental designs. Potential instruments might include historical zoning changes or hyper-local land-use regulations that affect the supply of Airbnb listings without directly influencing nightly rates. Furthermore, as the market matures, researchers could employ a Difference-in-Differences (DiD) approach to analyze the impact of specific regulatory shocks—such as the sudden implementation of registration requirements or short-term rental caps—on host pricing power across different boroughs.

Beyond endogeneity, there is a significant opportunity to explore the role of algorithmic pricing tools. As professionalization continues to reshape the platform, future studies should investigate how the adoption of dynamic pricing software alters the "Commoditization Trap," specifically determining if algorithmic intervention can help "mom-and-pop" hosts overcome the cognitive frictions and competitive pressures identified in this paper. Such inquiries would further clarify the boundaries between Monopolistic and Perfect Competition in the evolving digital hospitality landscape.

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Appendix

Robustness Checks

To check robustness of results, we rerun regression on the sub neighborhood competition. That is on the number of competitions situated in a town/city with the various New York City Neighborhoods. Results show robustness and consistency

Table 3: Regression Results: The Impact of Density on Airbnb Pricing-Robustness Checks

	<i>Dependent variable:</i>		
	Naive (1)	log(price) Controlled (2)	Interaction (3)
competitors_in_subhood	0.00008*** (0.00001)	−0.00008*** (0.00001)	−0.00006*** (0.00001)
dist_to_center_km		−0.05665*** (0.00101)	z −0.05692*** (0.00101)
room_typeHotel room	4.34359*** (0.04916)	4.18002*** (0.04599)	2.92666*** (0.09643)
room_typePrivate room	−0.79213*** (0.01045)	−0.64539*** (0.01010)	−0.59427*** (0.01500)
room_typeShared room	−0.85177*** (0.06375)	−0.63955*** (0.05964)	−0.27420*** (0.08869)
bedrooms	0.11306*** (0.00518)	0.15093*** (0.00488)	0.15182*** (0.00486)
competitors_in_subhood:room_typeHotel room			0.00104*** (0.00007)
competitors_in_subhood:room_typePrivate room			−0.00005*** (0.00001)
competitors_in_subhood:room_typeShared room			−0.00034*** (0.00006)
Constant	5.19468*** (0.01165)	5.65950*** (0.01369)	5.64009*** (0.01480)
Observations	21,242	21,242	21,242
R ²	0.43088	0.50400	0.51036
Adjusted R ²	0.43075	0.50386	0.51015

Note:

*p<0.1; **p<0.05; ***p<0.01

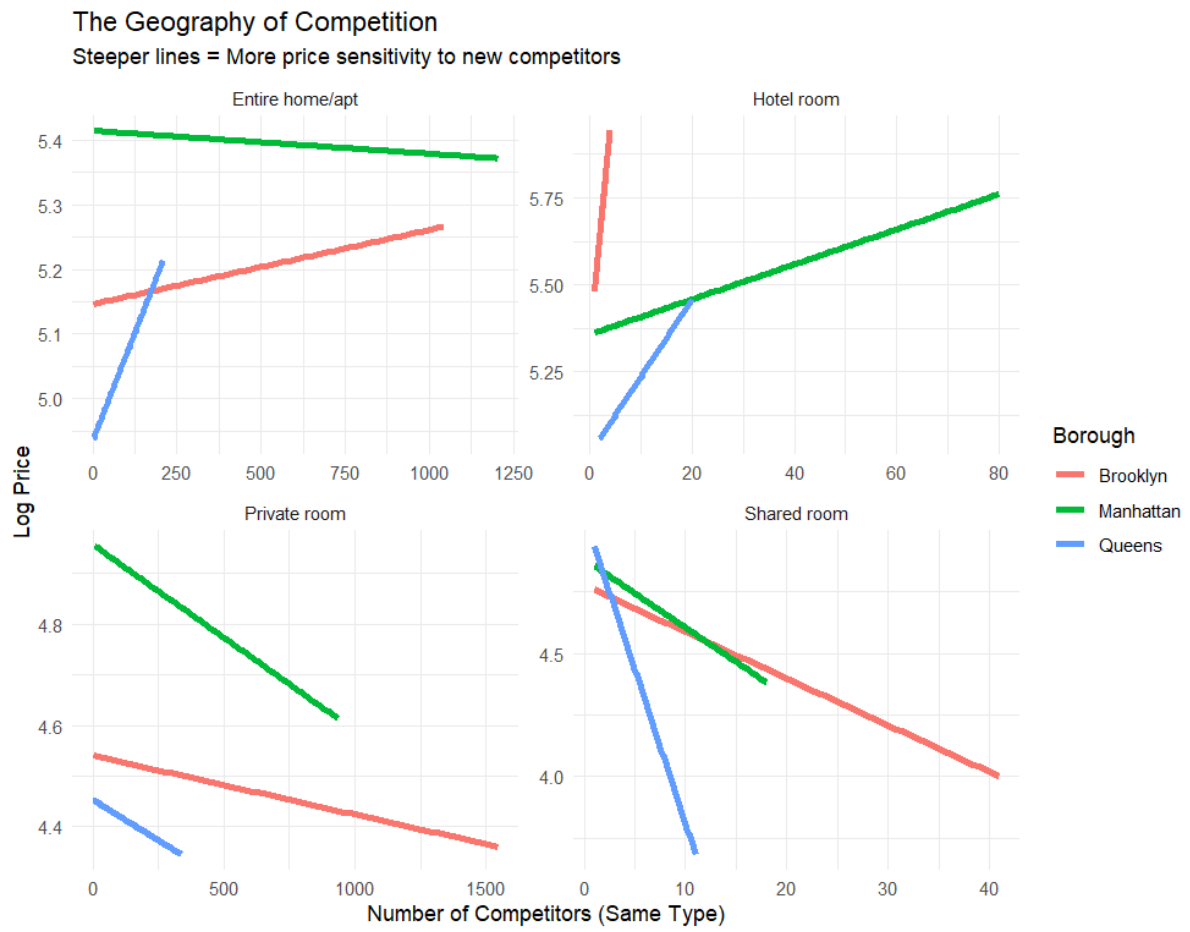


Figure 1: